AquaSense: Low-Cost Soil Moisture Monitoring with LoRa Sensor Networks for Small Farms

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Abstract

AquaSense is a low-cost soil moisture monitoring system that uses LoRa radio to send data over distances up to 5 kilometres. Each sensor node costs about 12 dollars and runs for 14 months on two AA batteries.

1 Introduction

Small farms often irrigate on a fixed schedule because commercial moisture sensors are expensive and need mobile network coverage.

We design a sensor system that a small farm can afford and that works in areas without mobile network.

2 Method

Each node contains a capacitive soil moisture sensor and a microcontroller that wakes up every 30 minutes to take a reading.

Readings are sent to a gateway using LoRa, and the gateway uploads them to a cloud dashboard where the farmer can see the moisture of each field.

3 Results

We deployed 20 nodes in a wheat field of 8 hectares for one growing season. The system reduced irrigation water use by 27 percent compared with the fixed schedule used by the farmer.

Packet loss was below 3 percent for nodes within 3 kilometres of the gateway.

4 Conclusion

AquaSense shows that affordable sensors can help small farms save water without mobile network coverage.

Future work includes adding solar panels to remove the need for battery changes.

References

[1] LoRa Alliance. LoRaWAN specification. 2020.

[2] FAO. Water use in agriculture. 2018.